

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Electronic Devices and Circuits

Pass Marks: 45

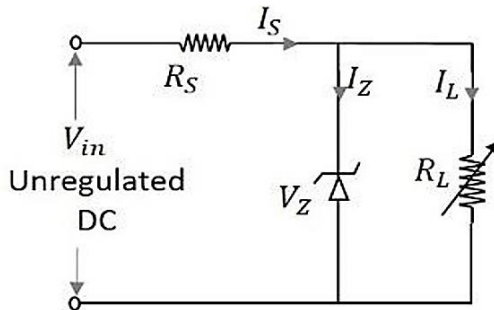
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

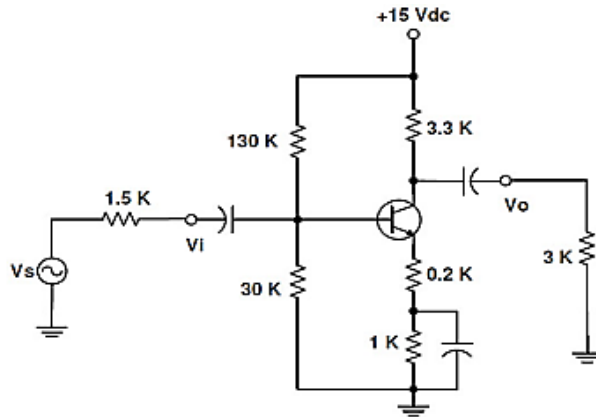
The figures in the margin indicate full marks.

Attempt all the questions.

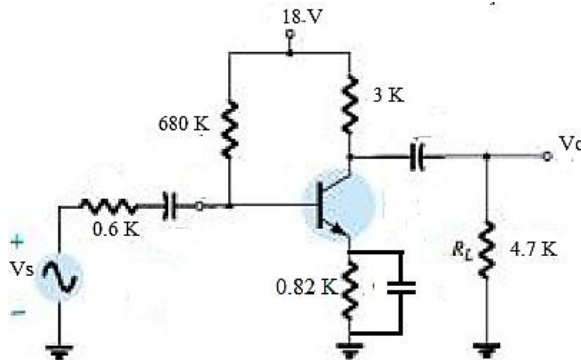
1. a) Explain the importance of Semiconductor materials on computer engineering. Explain the forward and reverse biasing of PN Junction diode with circuit diagram. 7
- b) If Zener voltage (V_Z) = 6V, Series resistance (R_S) = 200Ω and load resistance (R_L) = 750Ω . Find output voltage (V_o), Load current (I_L) and Zener Current (I_Z) if,
 - i. $V_{in} = 8\text{ V}$
 - ii. $V_{in} = 12\text{ V}$ 8



2. a) Explain input and output characteristics of common emitter configuration with detail necessary diagram. Also explain the concept of reach through and early effect. 8
 - b) For the circuit given below:
 - i. Draw DC load line
 - ii. Find Q point 7
- (Assume $\beta=90$, $V_{cc}=+15\text{ V}$)



3. a) Define regulated and unregulated power supplies. How can we use transistor series regulated power supply? Explain with necessary circuit diagram. 7
- b) Why is JFET called Unipolar Electronic Device? Describe the drain transfer characteristics of JFET with the help of necessary construction diagram and circuit diagram. 8
4. a) Find Overall voltage gain using r_e model. Assume $\beta = 100$ 8



- b) What are multistage amplifiers? Derive the equation for the overall voltage of multistage amplifier in terms of the individual voltage gains? 7
5. a) What is cross over distortion? Explain how this distortion is minimized in class AB amplifier with necessary diagrams. 7
- b) An amplifier has an input 25mV and a gain of 100 without the feedback. The distortion produced at the output of the amplifier is 10 %. It is desired to reduce the distortion to 3 % by using the 8

negative feedback. Calculate the gain and the output voltage with the feedback.

6. a) What are the characteristics of ideal Op-Amp? Design an operational amplifier using two inverting configurations to produce the output $V_0 = 10V_1 - 0.2V_2 + 5V_3 - 4V_4$. 7
- b) What are Barkhausen criterion for oscillation? Design Wein Bridge Oscillator that Oscillates about 10 kHz. 8
7. Write short notes on: (**Any two**) 2×5
- a) Importance of stability factor
- b) Clipper and clamping circuits
- c) Diode switching times